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Nease et al.

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(54) **CHAMFERED SINKER BAR CONNECTION**

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See application file for complete search history.

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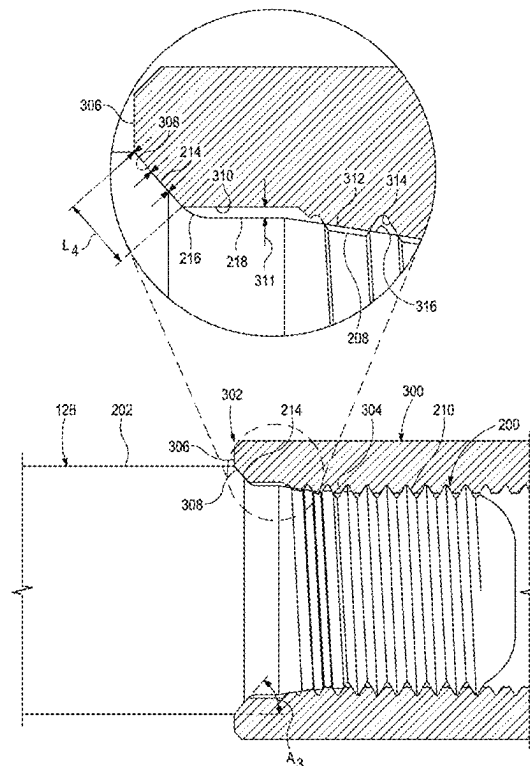
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(57) **ABSTRACT**

A sinker bar includes a cylindrical body having an outer surface terminating at a pin end. The pin includes a transition surface extending from an outer surface of the sinker bar and including a chamfered surface extending from the outer surface, an arcuate surface extending from the chamfered surface, and a straight-line surface extending from the arcuate surface, a tapered section extending from the straight-line surface, and a threaded section extending from the tapered section and terminating at a pin face.

20 Claims, 3 Drawing Sheets



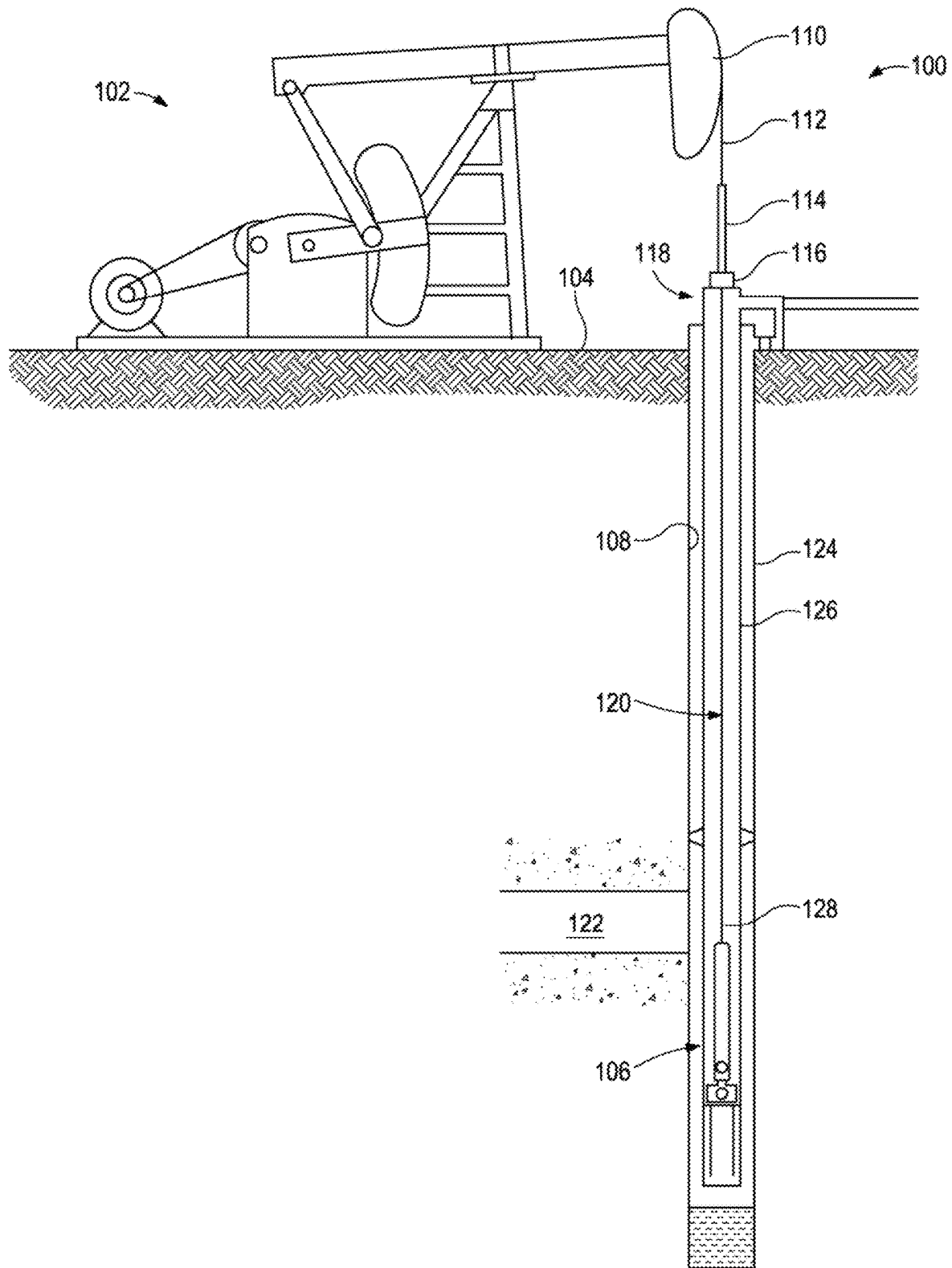


FIG. 1

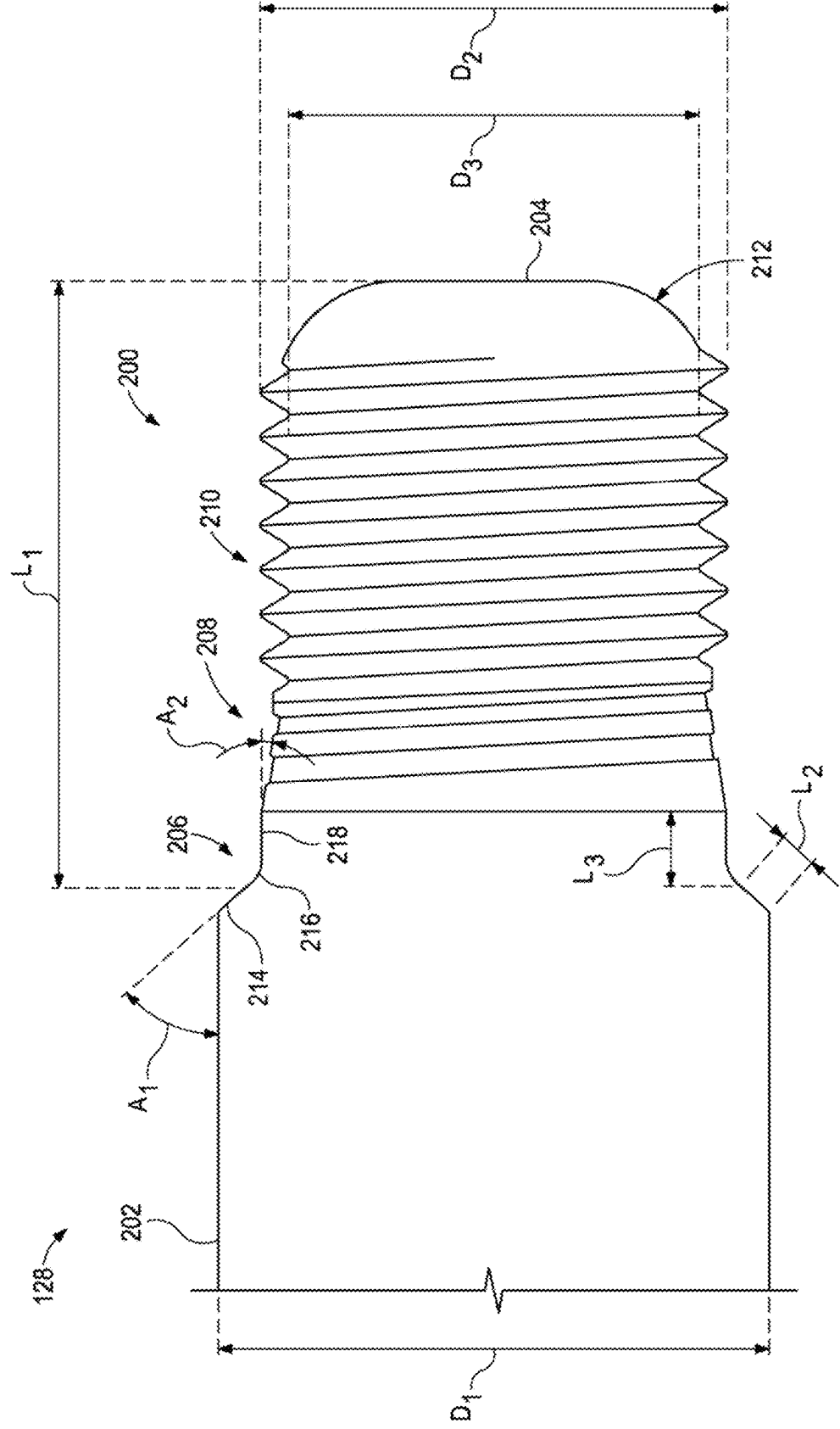


FIG. 2

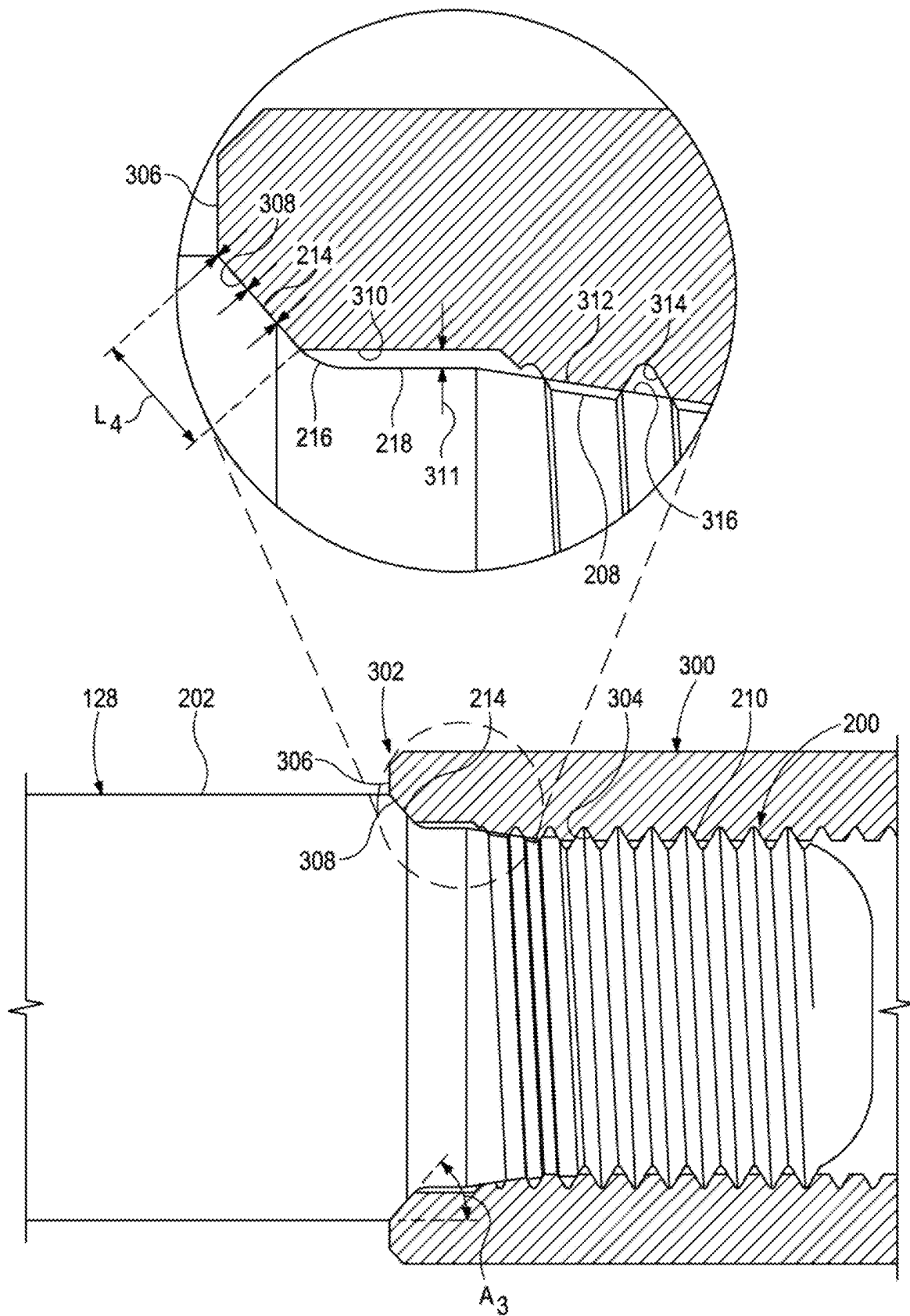


FIG. 3

CHAMFERED SINKER BAR CONNECTION

BACKGROUND

In the oil and gas industry, a sinker bar is a commonly utilized rod string component that is deployed to add weight to a rod string. Added weight may be desirable for any number of reasons not limited to overcoming downhole wellbore pressure, greater ease and efficiency in extending the rod string into a wellbore, as well as adding tension and or rigidity to the rod string. An operator may add any number of sinker bars that is procedurally desirable for the respective operation.

Typical sinker bars have opposing pin ends, thus the addition of sinker bars requires female-female couplings that interpose axially adjacent sinker bars. A successful coupling connection results in the greatest amount of surface contact between the pin end threads of the sinker bar and the internal threads of the coupling. In conventional sinker bar coupling configurations, the design of the pin end is sub-optimal, thus resulting in well operators over-torquing the coupling to the pin end, which frequently leads to cracked and failed couplings.

Accordingly, a connection that increases or maximizes the amount of surface contact between the external threads of the sinker bar and the internal threads of the coupling, and similarly increases the surface contact between the face of the pin and the face of the coupling, is desirable.

BRIEF DESCRIPTION OF THE DRAWINGS

The following figures are included to illustrate certain aspects of the present disclosure and should not be viewed as exclusive embodiments. The subject matter disclosed is capable of considerable modifications, alterations, combinations, and equivalents in form and function, without departing from the scope of this disclosure.

FIG. 1 is a schematic diagram of an example well system, according to one or more embodiments of the present disclosure.

FIG. 2 is a side view of a pin end of a sinker bar, according to one or more embodiments of the present disclosure.

FIG. 3 is a partial side view of a threaded engagement between the pin end of the sinker bar and an example coupling, according to one or more embodiments of the present disclosure.

SUMMARY OF THE DISCLOSURE

Various details of the present disclosure are hereinafter summarized to provide a basic understanding. This summary is not an extensive overview of the disclosure and is neither intended to identify certain elements of the disclosure, nor to delineate the scope thereof. Rather, the primary purpose of this summary is to present some concepts of the disclosure in a simplified form prior to the more detailed description that is presented hereinafter.

Embodiments of the present disclosure include a sinker bar that includes a cylindrical body having an outer surface terminating at a pin end. The pin end may include a transition surface extending from an outer surface of the sinker bar and including a chamfered surface extending from the outer surface, an arcuate surface extending from the chamfered surface, and a straight-line surface extending from the arcuate surface, a tapered section extending from the straight-line surface, and a threaded section extending from the tapered section and terminating at a pin face.

Embodiments of the present disclosure may further include a pin end of a rod string component, including a transition surface extending from an outer surface of the rod string component. The transition surface may include a chamfered surface extending from the outer surface, an arcuate surface extending from the chamfered surface, and a straight-line surface extending from the arcuate surface.

The pin end may further include a tapered section extending from the straight-line surface, and a threaded section extending from the cone section and terminating at a pin face.

Embodiments of the present disclosure may further include a method of making-up a pin end of a sinker bar, the method may include advancing the pin end into an interior of a coupling that defines internal threading, the pin end extending from an outer surface of a cylindrical body of the sinker bar and including a transition surface extending from an outer surface of the sinker bar and including a chamfered surface extending from the outer surface, an arcuate surface extending from the chamfered surface, and a straight-line surface extending from the arcuate surface, a tapered section extending from the straight-line surface, and a threaded section defining external threading and extending from the tapered section and terminating at a pin face. The method may further include rotating at least one of the pin end and the coupling and thereby threadably mating the internal and external threading.

DETAILED DESCRIPTION

Embodiments in accordance with the present disclosure generally relate to sinker bars used in the oil and gas industry and, more particularly, to improved pin end designs and threading for sinker bars.

Embodiments disclosed herein describe sinker bars having a pin end connection operable to couple to a universal coupling, where the connection is matable as designed and there exists little to no space (or “gap”) between the shoulder of the pin end connection and the coupling. The pin end designs disclosed herein are configured to maximize the surface contact area between the external threads of the pin end and the internal threads of a receiving coupling. The presently disclosed pin end designs can maximize surface face contact between the face of the pin and the face of the coupling when the sinker bar and the coupling are connected and torqued.

The pin end designs described herein are compatible with a $\frac{3}{4}$ " (0.75 inch) polished rod pin end that has been implemented in sinker bars having an outer diameter of $1\frac{1}{4}$ " (1.25 inches). Polished rods and their connections are subject to the dimensional tolerances specified by the governing industry/regulatory body, the American Petroleum Institute (API). Accordingly, the pin end designs disclosed herein abide by the applicable API requirements and simultaneously provide for an optimal connection between the sinker bar and a coupling without over-torquing, and thus potentially damaging the coupling.

FIG. 1 is a schematic view of an example well system **100** that may employ the principles of the present disclosure. The well system **100** includes a pumping unit **102** positioned at a well surface **104**. The pumping unit **102** is known in the industry by many names, including, but not limited to, a “pumpjack,” a “nodding donkey pump,” or a “horsehead pump”, and is configured to drive (operate) a submersible pump **106** arranged within a wellbore **108**. As illustrated, the pumping unit **102** includes a horse head **110** to which a cable **112** (alternately referred to as a “bridle”) is attached. A

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polished rod **114** is attached to the distal end of the cable **112** and extends through a stuffing box **116** forming part of a wellhead **118** that caps the wellbore **108**. The polished rod **114** is reciprocated within stuffing box **116** as the horse head **110** moves (reciprocates) up and down.

The polished rod **114** forms part of or is otherwise operatively coupled to a rod string **120** extended into the wellbore **108**.

As shown, the wellbore **108** extends vertically into a subterranean formation and penetrates a hydrocarbon-bearing reservoir **122**. The wellbore **108** may be lined with casing **124** that extends to (or toward) the bottom (most distal end) of the wellbore **108**. An extension of tubing **126** extends from the wellhead **118** and is concentrically arranged within the casing **124**. The tubing **126** terminates at a predetermined distance from the bottom of the wellbore **108**.

The rod string **120** may include a variety of components that enable pumping of hydrocarbons migrating into the wellbore **108** to the well surface **104** using the pumping unit **102**. For example, the rod string **120** can include the submersible pump **106** arranged at the distal end of the rod string **120** and one or more sinker bars **128**. The sinker bar **128** comprises a weighted component made of a high-density material and used to maintain tension in the rod string **120**. When used in a slickline operation, the sinker bar **128** may help overcome the effects of pressure and friction caused by the wellhead **118**.

The sinker bar **128** may be operatively coupled to the submersible pump **106**, either directly or indirectly. In some applications, one or more stabilizing components (not shown) may be included in the rod string **120** and interpose the submersible pump **106** and the sinker bar(s) **128**. The sinker bar(s) **128** may be arranged within and operatively coupled to any matable component within the rod string **120**.

Those of ordinary skill in the art will be familiar with polished rods. Generally, a polished rod is the uppermost component of the rod string **120**. The API sets forth requirements and acceptable tolerances for polished rods and their respective connections. Although specifically designed for polished rods, over time, the oil and gas industry has adopted a $\frac{3}{4}$ " (0.75 inch) polished rod specific connection, and implemented the connection in $1\frac{1}{4}$ " (1.25 inch) sinker bars, for which the connection was not designed.

Sinker bars (e.g., the sinker bar **128**) are generally manufactured so that opposing ends of the sinker bar terminate at a pin end connection that includes a radial shoulder that extends substantially perpendicular to the longitudinal axis of the sinker bar. Accordingly, the radial shoulder of conventional sinker bars commonly comprises a 90° transition to the pin end connection. Depending on the design of the coupling used to mate with the sinker bar pin end connection, the radial shoulder can cause issues when mating (making up) the $1\frac{1}{4}$ " sinker bar with the $\frac{3}{4}$ " polished rod connection to a coupling, for the purposes of creating the rod string **120** extended into the wellbore **108**.

Sinker bar couplings are often manufactured with an internal bevel that extends from the face of the coupling, as described in more detail below with reference to FIGS. 2 and 3. Internal threads are defined within the interior of the coupling at a predetermined distance from the bevel. Generally, the bevel exhibits an angle of either 45° or 30° . Because of the API specified dimensions for the $\frac{3}{4}$ " polished rod pin end connection, upon make-up, the 90° shoulder prevents full engagement between the sinker bar threads (comprising the polished rod pin end connection) and the internal threads of the coupling. Instead, the configuration

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causes the threads to "lock up" prior to fully engaging the coupling. Such lock up creates a gap between the uppermost portion of the coupling (the face) and the radial shoulder of the sinker bar **128**.

In an effort to eliminate the gap, rig site operators will over-torque the connection, which oftentimes results in cracking or damaging the coupling. In the alternative, the inverse can occur during make-up, where the shoulder of the sinker bar pin bottoms out on the shoulder of the coupling before the internal tapered area of the threaded portion of the pin end is able to properly seat. When this happens, galling occurs on the face of the coupling and also on the face of the sinker bar.

According to embodiments of the present disclosure, this issue may be mitigated, or eliminated entirely, via a re-design of the $1\frac{1}{4}$ " sinker bar **128** featuring a $\frac{3}{4}$ " polished rod connection, as disclosed herein. Although, the present disclosure pertains to a re-design of the $1\frac{1}{4}$ " sinker bar **128** with a $\frac{3}{4}$ " polished rod connection, other embodiments may comprise a sinker bar **128** having an outer diameter of $1\frac{3}{8}$ ", $1\frac{1}{2}$ ", $1\frac{5}{8}$ " $1\frac{3}{4}$ " or 2". Similarly, the embodiments disclosed herein may also comprise polished rod connections including $\frac{3}{4}$ ", $\frac{7}{8}$ " or 1".

FIG. 2 is an enlarged side view of an example pin end connection or "pin end" **200** of the sinker bar **128** of FIG. 1, according to one or more embodiments of the present disclosure. As illustrated, the sinker bar **128** comprises a cylindrical solid body having a generally smooth outer surface **202** that exhibits an outer diameter D1. In at least one embodiment, the outer diameter D1 may be about $1\frac{1}{4}$ " (1.25 inches), but may be more or less than $1\frac{1}{4}$ ", as governed by API required tolerances (i.e., $+0.005$, -0.010).

The sinker bar **128** terminates with the pin end **200**, which exhibits a pin length L1. In some applications, the pin length L1 may accord with API standard pin lengths. In such embodiments, for instance, the pin length L1 may be about 1.375 inches, but could alternatively be slightly longer or shorter depending on the application or resulting from manufacturing tolerances.

The pin end **200** includes multiple features that ultimately terminate at a pin face **204**, which constitutes the distal-most point of the sinker bar **128**. As illustrated, the pin end **200** may include a transition profile **206** that transitions from the outer surface **202**, a tapered section **208** that transitions from the transition profile **206**, external threading **210** that transitions from the tapered section **208**, and a pin face arcuate surface **212** that provides a transition from the external threading **210** to the pin face **204**.

The external threading **210** may be configured to mate with internal threading defined on a coupling (not shown). In such embodiments, the external threading **210** and the corresponding internal threading of the coupling may comprise matable, helical threads. Operatively coupling and threading the pin end **200** to a coupling enables the rod string **120** (FIG. 1) to be extended into the wellbore **108** (FIG. 1). To ensure proper make-up of the sinker bar **128** to a coupling, it is advantageous that the respective threads are both engaged and torqued as designed. As illustrated, the external threading **210** includes a major diameter D2, which may range between about 1.0482 inches and about 1.0612 inches, and a minor diameter D3, which may be about 0.9384 inches. The pin face arcuate surface **212** may exhibit an arc length radius of about 0.2810 inches. As will be appreciated, however, the foregoing dimensions may be altered depending on the application.

In the illustrated embodiment, the transition profile **206** may define or otherwise provide multiple contiguous sur-

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faces that provide a transition between the outer surface **202** of the sinker bar **128** at the outer diameter **D1** to the tapered section **208**. In at least one embodiment, as illustrated, the transition profile **206** may include a straight or “chamfered” surface **214** that extends from the outer surface **202**, an arcuate surface **216** that extends from the chamfered surface **214**, and a straight-line surface **218** that extends from the arcuate surface **216**.

The chamfered surface **214** extends from the outer surface **202** at an angle **A1**, which, in at least one embodiment, may be 45°. In other embodiments, however, and in accordance with API recommendations, the angle **A1** may be that which is operationally feasible and desirable to the manufacturer. The chamfered surface **214** may extend between the outer surface **202** and the arcuate surface **216** over a length **L2**, which may be about 0.0738 inches.

The chamfered surface **214** transitions directly into the arcuate surface **216**, which may exhibit an arc length radius of about 0.0938 inches. The arcuate surface **216** transitions directly into the straight-line surface **218**. In some embodiments, the straight-line surface **218** extends parallel to the outer surface **202** and otherwise parallel to the central axis of the sinker bar **128**. In at least one embodiment, the arcuate surface **216** and the straight-line surface **218** may exhibit a combined axial length **L3** of about 0.175 inches before transitioning to the tapered section **208**. In other embodiments, the length **L3** may be greater than 0.175 inches but less than about 0.225 inches. In yet another embodiment, the length **L3** may be 0.225 inches in length. Accordingly, in at least one embodiment, the transition profile **206** includes three contiguous and continuous surfaces, including the chamfered surface **214**, the arcuate surface **216**, and the straight-line surface **218**.

The tapered section **208** may extend from the straight-line surface **218** at an angle **A2**, which may be about 9°, as specified by API requirements. In some embodiments, as illustrated, the external threading **210** may extend into the tapered section **208**, but at a reduced diameter. In other embodiments, however, the external threading **210** may stop before entering the tapered section **208**. In such embodiments, the tapered section **208** may exhibit a substantially smooth outer surface.

While the pin end **200** is described herein as forming part of the sinker bar **128**, it is contemplated herein that the pin end **200** may alternatively be implemented in any downhole rod string connection. Accordingly, manufacture of the pin end **200** is not limited to the sinker bar **128**, and implementation in a component other than a sinker bar or polished rod does not exceed the scope of this disclosure.

FIG. 3 is a partial side view of a threaded engagement between the pin end **200** and an example coupling **300**, according to one or more embodiments of the present disclosure. FIG. 3 also provides an enlarged inset graphic of a portion of the threaded engagement.

As illustrated, the pin end **200** is sized to be received within the coupling **300** at one end **302** of the coupling **300**. The interior of the coupling **300** defines or otherwise provides internal threading **304** configured to threadably mate with the external threading **210** of the pin end **200**. The end **302** of the coupling **300** provides a face **306**, which comprises a generally flat surface extending substantially perpendicular to the outer surface **202** of the sinker bar **128**. The face **306** transitions to an internal bevel **308** extending from the face **306** at an angle **A3**. In some embodiments, the angle **A3** may be the same as the angle **A1** (FIG. 2), thereby enabling mating engagement between the internal bevel **308** and the chamfered surface **214**. In such embodiments, the

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angles **A1**, **A3** may each be about 45°, which allows the internal bevel **308** to make contact with the entire length **L2** (FIG. 2) of the chamfered surface **214**, as indicated by the opposing angles provided in the enlarged inset graphic.

In other embodiments, however, the angle **A3** of the internal bevel **308** may be 30°, without departing from the scope of the disclosure. As those of ordinary skill will be familiar, 45° and 30° machined angle(s) **A3** are considered to be standard. However, the scope of this disclosure is not intended to limit the angle **A3** of the internal bevel **308**. Accordingly, a manufacturer may machine another angle **A3** at their discretion without exceeding the scope of this disclosure.

Because the sinker bar **128** includes the chamfered surface **214**, as opposed to a traditional 90° radial shoulder, the chamfered surface **214** is able to align and make significant surface contact with the internal bevel **308** of the coupling **300**. This may prove advantageous in enabling an increased amount of surface contact between the internal bevel **308** and the chamfered surface **214**, as compared to a sinker bar with a conventional 90° radial shoulder.

As best seen in the enlarged inset graphic, the internal bevel **308** exhibits a length **L4**, which may be greater than the length **L2** of the chamfered surface **214**. In the illustrated embodiment, for example, the length **L4** of the internal bevel **308** extends into the arcuate surface **216** of the transition profile **206** (FIG. 2) for the pin end **200**. The internal bevel **308** may then transition into a straight-line section **310** that extends substantially parallel to the straight-line surface **218** of the transition profile **206**. In at least one embodiment, as illustrated, a gap **311** may be defined between the straight-line section **310** and the straight-line surface **218** when the coupling **300** is fully threaded to the pin end **200**.

The straight-line section **310** may then transition to an internal cone section **312** configured to matably engage the tapered section **208** of the pin end **200**. In some embodiments, the cone section **312** may extend at an angle substantially similar to the angle **A2** (FIG. 2) of the tapered section **208**. In some embodiments, as illustrated, the cone section **312** may define internal threading **314**. In embodiments where the tapered section **208** includes external threading **316**, as illustrated, the internal threading **314** may be configured to threadably mate with the external threading **316**.

The configuration described herein allows for threaded engagement between the sinker bar **128** and the coupling **300** while simultaneously allowing for proper surface contact between the internal bevel **308** of the coupling **300** and the chamfered surface **214** of the pin end **200**. By providing substantial engagement between the opposing chamfered surface **214** and the internal bevel **308**, and threaded engagement between the external and internal threading **210**, **304**, proper torque may be applied by the operator to make-up the coupling **300**. This may eliminate any gap between the face **306** of the coupling **300** and the transition profile **206** (FIG. 2) of the sinker bar **128** and, as a result, mitigating damage to the coupling **300** during make-up (i.e., threading the coupling **300** onto the pin end **200**).

Alternatively, due to design tolerances (i.e., tolerance stack-up), there may be instances when the tapered section **208** does not exactly (or closely) align with the corresponding threads of the coupling **300**. In such instances, because the sinker bar **128** includes the chamfered surface **214** as opposed to a traditional 90° radial shoulder, adequate surface contact is still achievable between the internal bevel **308** of the coupling **300** and the chamfered surface **214** of

the sinker bar **128**. This configuration still allows adequate torque, within API tolerances, to be applied and achieved.

As used herein, the phrase “at least one of” preceding a series of items, with the terms “and” or “or” to separate any of the items, modifies the list as a whole, rather than each member of the list (i.e., each item). The phrase “at least one of” allows a meaning that includes at least one of any one of the items, and/or at least one of any combination of the items, and/or at least one of each of the items. By way of example, the phrases “at least one of A, B, and C” or “at least one of A, B, or C” each refer to only A, only B, or only C; any combination of A, B, and C; and/or at least one of each of A, B, and C.

The use of directional terms such as above, below, upper, lower, upward, downward, left, right, uphole, downhole and the like are used in relation to the illustrative embodiments as they are depicted in the figures, the upward direction being toward the top of the corresponding figure and the downward direction being toward the bottom of the corresponding figure, the uphole direction being toward the surface of the well and the downhole direction being toward the toe of the well.

What is claimed is:

1. A sinker bar, comprising:
 - a cylindrical body having an outer surface terminating at a pin end, the pin end including:
 - a transition surface extending from an outer surface of the sinker bar and including a chamfered surface extending from the outer surface, an arcuate surface extending from the chamfered surface, and a straight-line surface extending from the arcuate surface;
 - a tapered section extending from the straight-line surface; and
 - a threaded section extending from the tapered section and terminating at a pin face,
 wherein the tapered section axially interposes the straight-line surface and the threaded section.
2. The sinker bar of claim 1, wherein the chamfered surface extends from the outer surface at an angle of 45 degrees.
3. The sinker bar of claim 1, wherein the chamfered surface extends from the outer surface at an angle of 30 degrees.
4. The sinker bar of claim 1, wherein the outer surface exhibits a diameter of 1¼ inches.
5. The sinker bar of claim 1, wherein the outer surface exhibits a diameter selected from the group consisting of 1¼ inches, 1⅜ inches, 1½ inches, 1⅝ inches, 1¾ inches, and 2 inches.
6. The sinker bar of claim 1, wherein the threaded section defines external threads threadably matable with internal threads defined within a coupling.
7. The sinker bar of claim 6, wherein the coupling defines an internal bevel engageable with the chamfered surface.
8. A pin end of a rod string component, comprising:
 - a transition surface extending from an outer surface of the rod string component, the transition surface including:
 - a chamfered surface extending from the outer surface;
 - an arcuate surface extending from the chamfered surface; and
 - a straight-line surface extending from the arcuate surface;

a tapered section extending from the straight-line surface; and

a threaded section extending from the tapered section and terminating at a pin face,

wherein the tapered section axially interposes the straight-line surface and the threaded section.

9. The pin end of claim 8, wherein the threaded section defines external threads that are threadably matable with internal threads defined by a coupling.

10. The pin end of claim 8, wherein the chamfered surface extends from the outer surface at an angle of 45°.

11. The pin end of claim 8, wherein the chamfered surface exhibits a length of about 0.0738 inches.

12. The pin end of claim 8, wherein the rod string component exhibits a diameter of about 1¼ inches.

13. The pin end of claim 8, wherein the tapered section extends from the straight-line surface at an angle of 9°.

14. The pin end of claim 8, wherein the rod string component comprises a sinker bar.

15. A method of making-up a pin end of a sinker bar, the method comprising: advancing the pin end into an interior of a coupling that defines internal threading, the pin end extending from an outer surface of a cylindrical body of the sinker bar and including:

- a transition surface extending from an outer surface of the sinker bar and including a chamfered surface extending from the outer surface,

- an arcuate surface extending from the chamfered surface, and

- a straight-line surface extending from the arcuate surface;

- a tapered section extending from the straight-line surface; and

- a threaded section defining external threading and extending from the tapered section and terminating at a pin face, wherein the tapered section axially interposes the straight-line surface and the threaded section; and

- rotating at least one of the pin end and the coupling and thereby threadably mating the internal and external threading.

16. The method of claim 15, wherein an end of the coupling provides a face extending perpendicular to the outer surface of the sinker bar, and the face transitions to an internal bevel extending from the face at an angle offset from perpendicular, wherein rotating the at least one of the pin end and the coupling comprises rotating the at least one of the pin end and the coupling until the internal bevel engages against the chamfered surface.

17. The method of claim 16, wherein the chamfered surface and the internal bevel each exhibit an angle of 45 degrees.

18. The method of claim 16, wherein the internal bevel exhibits an angle of 30 degrees.

19. The method of claim 16, wherein the internal bevel exhibits a length greater than a length of the chamfered surface.

20. The method of claim 16, wherein the internal bevel transitions into a straight-line section extending substantially parallel to the straight-line surface, and wherein a gap is defined between the straight-line section and the straight-line surface.

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